Machine Learning Stellar Classification

Rediscovering the HR Diagram

Krupa Pothiwala

Florida Institute of Technology

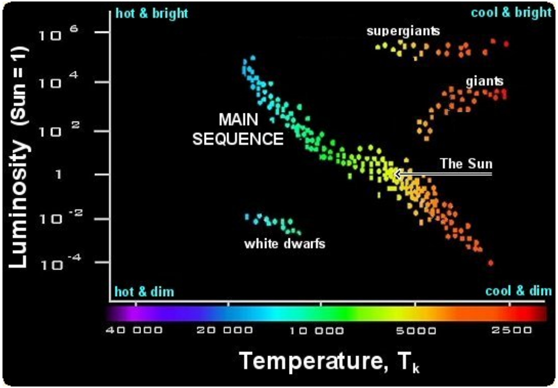
May 2026

Outline

The Hertzsprung-Russell Diagram

The HR diagram maps where stars are in their **life cycle**:

- **Main Sequence** - where most stars (including our Sun) spend their lives
- **Giants & Supergiants** - post-main-sequence phase
- **Turn-off point** - stars diverge as core hydrogen runs out
- **White Dwarfs** - stellar remnants in the lower-left



Conventional Definition

There are certain terms that would be useful to know going forward

- mas - milli arcsecond (angular measurement). Take a full circle of 360 degrees and one degree is 60 arcminutes so one arcsecond is 1/3600th of degree.
- G band - G filter on the telescope, measures white light $(330 - 1050nm)$
- parsec (ϖ) - unit of length used in astronomy. 3.26 light years $\approx$ 1 parsec
- parallax - distance measurements (mas). Formula used for it is parsecs $= 1000/\varpi$.
- apparent magnitude (m) - brightness as it appears to us. Depends on the band.
- Absolute magnitude (M) - brightness of the object if it was 10 parsecs away from Earth. Normalized magnitude. Depends on the band
- Extinction (A) - how much light gets absorbed between us and the object. Depends on the band.

Research Question

Can unsupervised machine learning **rediscover** the structure of the HR diagram from raw, unlabelled stellar data *with no prior knowledge of stellar physics?*

Data: Gaia DR3, NGC 2516

Source: ESA Gaia Data Release 3 European Space Agency (2022b) and European Space Agency (2022a)

Retrieved via `astroquery.gaia` (Python) Ginsburg et al. (2023)

Cone Search Parameters:

- RA $= 119.517^\circ$, Dec $= -60.752^\circ$
- Radius: 1.0°
- Parallax: 1.5–3.5 mas
- $\mu_\alpha = -4.7 \pm 3.0$ mas/yr
- $\mu_\delta = 11.2 \pm 3.0$ mas/yr
- Parallax error < 0.5 mas

Columns Retrieved:

- Apparent magnitudes (G, BP, RP)
- Color indices
- Proper motions
- Parallax
- Extinction estimates

Final Sample

2,670 member stars

Data Cleaning & Feature Engineering

Cleaning Steps:

1. Drop rows missing m_G, ϖ, BP$-$RP, μ
2. RUWE quality cut: remove RUWE ≥ 1.4 Pearce (2022)
3. Drop columns with $> 60\%$ missing values
4. Remove GSP-Phot (General Stellar Parametrizer from Photometry) columns (too incomplete).

Derived Features:

Absolute Magnitude:

$$M_G = m_G - 5\log_{10}(d) + 5 - A_G$$

where $d\,[\mathrm{pc}] = 1000/\varpi$

Corrected Color Index:

$$(BP - RP)_0 = (BP - RP) - E(BP - RP)$$

Removes interstellar reddening.

Machine Learning Pipeline

Features (4 inputs)

Apparent magnitudes: m_G, m_{BP}, m_{RP} $+$ Parallax: ϖ

1. **Z-score Normalisation**: rescale each feature to mean 0, std 1
2. **PCA**: compress 4 features into 2 principal components
3. **UMAP + HDBSCAN**: explored, but discarded

Why not UMAP + HDBSCAN?

HDBSCAN traced UMAP's layout, not real data structure; **pipeline bias**. PCA alone was sufficient.

Results: PCA Output

99.7% of total variance captured in 2 components

PC1 - Brightness Gradient

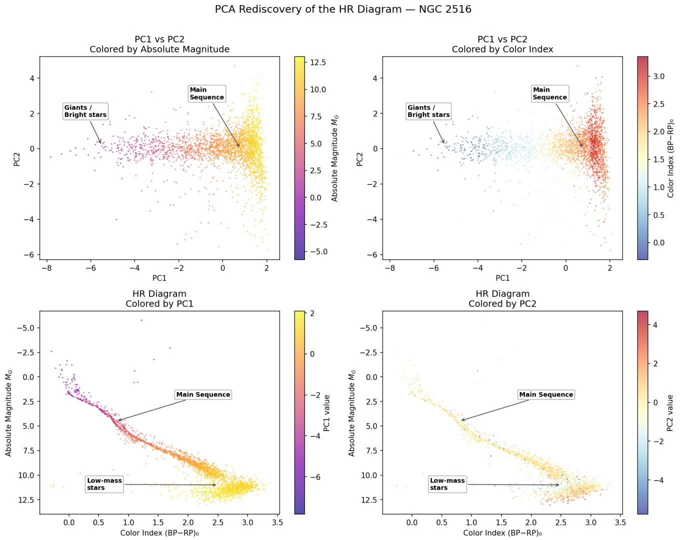
- Traces the full main sequence
- Hot blue stars at one end, faint red stars at the other
- Giants separate naturally at negative PC1

PC2 - Residual Scatter

- Near-constant along the main sequence
- Picks up parallax depth / cluster spread
- No new astrophysical information

The model recovered HR diagram structure **entirely without supervision**.

Result Visualization



Conclusion

- PCA recovered the HR diagram from raw photometric data with **no prior physics knowledge**
- 2 components captured **99.7%** of variance; almost no information lost
- PC1 $\equiv$ brightness / main sequence axis; PC2 $\equiv$ residual scatter
- UMAP + HDBSCAN introduced pipeline bias - complexity is not always better

Broader Insight

Stellar physics is structured enough that a simple linear model can rediscover fundamental astronomical relationships from scratch, given the right raw data.

References

European Space Agency (2022a). Gaia Archive: Extract Data. https://www.cosmos.esa.int/web/gaia-users/archive/extract-data. Accessed: 2026.

European Space Agency (2022b). Gaia Data Release 3 (DR3). https://www.cosmos.esa.int/web/gaia/dr3. Accessed: 2026.

Ginsburg, A. et al. (2023). Astroquery: Gaia Module Documentation. https://astroquery.readthedocs.io/en/latest/gaia/gaia.html. Accessed: 2026.

Pearce, L. (2022). RUWE as an Indicator of Multiplicity. http://www.loganpearcescience.com/research/RUWE_as_an_indicator_of_multiplicity.pdf. A review of Gaia Renormalized Unit Weight Error as a sensitivity indicator for unresolved multiple systems.